

COURSE

OVERVIEW

CLIMBING EQUIPMENT INSPECTION & MANAGEMENT

The Climbing Equipment Inspection and Management course provides learners with knowledge to inspect and manage work-at-height equipment safely. The course focuses on inspection regimes, traceability, storage, care, and rejection criteria.

Learners are introduced to legal standards, inspection frequencies, and responsibilities. Emphasis is placed on defect identification and record keeping.

BY COMPLETING THIS COURSE, LEARNERS WILL BE ABLE TO:

- Explain equipment management requirements
- Identify work-at-height equipment and components
- Apply inspection regimes
- Identify defects and rejection criteria
- Maintain inspection records and traceability
- Ensure equipment is stored and withdrawn correctly



INTRODUCTION

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Preface to the learner

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GRAVITY

VALUES



R

RELATIONSHIPS

Creating and maintaining meaningful relationships with all stakeholders of Gravity.



I

INTEGRITY

Conducting day-to-day business with integrity at all times.



S

SOLUTIONS **DRIVEN**

Being solution driven.



Q

QUALITY & **EXCELLENCE**

Striving for excellence and constant improvement.

WELCOME, LEARNER!

Working at height is inherently risky — and this includes training exercises. The best way to stay safe is by learning and practising the correct techniques. This manual will help you:

- Understand the basics of fall arrest and basic rescue
- Learn how to identify and manage height-related hazards
- Build the skills needed to work safely and confidently

Working at height means doing any job where you could fall from one level to another. This includes tasks done:

- Above the ground or floor
- Near open edges
- Near fragile surfaces
- Next to holes or gaps in the ground or floor
- Close to excavations

Note: Work at height does not include slipping or tripping hazards on the same level.

HOW TO USE THIS MANUAL?

This guide supports the Fall Arrest and Basic Rescue Course and is meant to be used with the help of a trained instructor.

At Gravity Training, all facilitators are highly trained to deliver top-quality learning experiences. The content has been designed to be clear, easy to remember, and practical for real worksites.

Before any work begins, the site must be safe.

Use this simple checklist

Follow the Rules:

Make sure you follow all client rules.

Example: If night work is not allowed, do not work at night.

01



02



Rescue Plan

Have a rescue plan ready. A rescue kit and first aid kit must be on site.

Work in Pairs

There must be at least two workers on site.
If something goes wrong, one can help or call for help.

03



05

Safe Site

Check the building or structure.

Make sure it is strong and safe to work on.



Check Equipment

All tools and gear must be safe and working before use.

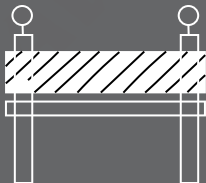
04



06 Manage Traffic

If there are cars or trucks nearby, there must be a traffic safety plan.

06

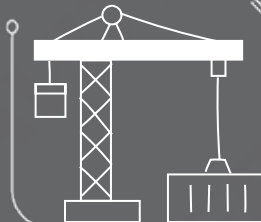


07 Do a Risk Assessment

Check what dangers there are.

All workers must understand the risks before starting work.

07



08

08 Use the Right Safety Gear

Make sure the fall protection fits the job and the site.

EXCLUSION ZONES & BARRICADES

Exclusion Zone: An area where no one is allowed to walk or work.

Example: A 2-metre space around a hole in the roof.

Barricade: A physical barrier to block people from danger.

Example: A fence or tape to stop people from falling.

SITE SAFETY



Be healthy and ready before starting work.



Only do jobs you are trained and allowed to do.



Never work under the influence of alcohol or drugs.



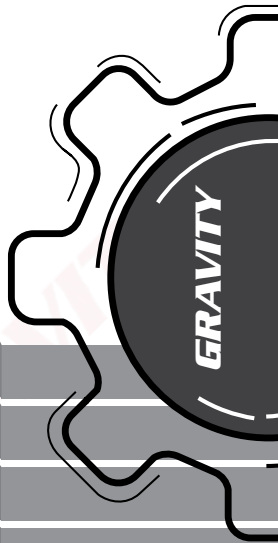
Always wear your safety gear (PPE, harness if working at height).



Use harnesses correctly and stay connected at heights.



Never work alone and always follow safety steps.



SITE SAFETY

Keep people away from areas below when working above.



Don't walk under lifted loads or stand in dangerous areas.



Only trained people can do electrical, underground, or confined space work.



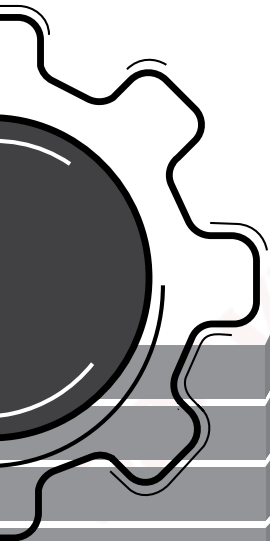
Follow road rules. Wear helmets. No passengers on open trucks..



Don't drive tired or distracted.



Stop unsafe work! Everyone has the right to speak up.



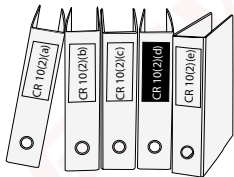


A. Why is a course in Climbing Equipment and Inspection Management necessary?

1. Legal requirements:

When examining the legal requirements, the fall protection plan must always be taken into consideration. In Construction Regulation 10 of the Occupational Health and Safety Act (OHS Act) it is stated that a contractor must appoint a person to create and maintain a fall protection plan, and within that plan the following must be included:

- CR10(2)(a) - Risk assessment: The purpose of the fall protection plan is to address and mitigate or control all fall risks, thus the risk assessment forms a crucial part of the fall protection plan.
- CR10(2)(b) - The process of evaluating the **medical fitness** of employees: All workers must be in possession of a valid certificate of fitness that must be renewed yearly.
- CR10(2)(c) - A training programme: All workers on site must receive the necessary training, and must be competent to perform the relevant task and to work on the relevant site..
- **CR10(2)(d) - Equipment management: A procedure must be in place for the inspection, testing and maintenance of climbing equipment. Equipment must be safe for use and fit for purpose on site.**
- CR10(2)(e) - Rescue plan: There must be an adequate rescue plan on site that can be performed at any time. This plan must stipulate the necessary procedure, personnel and equipment needed to rescue a fallen person.



We will only concern ourselves with CR10(2)(d): equipment management, which sets out the requirements that one needs to comply with when managing equipment. However, this regulation does not give guidance on how to do inspections, testing and maintenance on the equipment. For this reason, we use the International Organization for Standardization (ISO) as further guidance.

2. Standards for equipment inspection:

ISO 22846-2 Section 7 and the (OHS Act) place certain duties on the employers concerned with managing or carrying out work at height. The standard describes the procedures that are required to be followed in order to ensure that equipment used is safe for use and fit for purpose.

One also needs to refer to the original equipment manufacturer requirements (EN 365). This EN standard sets out the general requirements for instructions on use, maintenance, periodic examination, repair, marking and packaging that the manufacturer is required to follow.

Standards and legal requirements:

- Identify and select equipment that conforms to standards that are relevant to the intended use.
- Ensure that all equipment meets the relevant European standard (EN) / Compliance to European health, safety, and environmental protection requirements (CE Markinking)

European standard (EN) is a technical standard designed to be used as a rule or guideline. It is created by manufacturers, consumers and regulators of a product to ensure that a product is tested and approved for quality and safety. Every unique EN-standard item has its own testing criteria that it needs to comply with. The EN -standard is used in 34 countries.

Compliance with European health, safety, and environmental protection requirements (CE marking). All products manufactured in Europe with the CE marking signify that the product has been assessed to meet top-level health and environmental protection requirements. It is the responsibility of the manufacturer to ensure that the product complies with the CE standard before the product is sold.

- Ensure that equipment is safe for use and fit for intended purpose.
- Ensure that only a competent person conducts the detailed/interim inspections.
- Ensure that the venue in which the inspection is to be done is sufficiently spacious and lit.
- Ensure that a user performs a pre-use inspection on the equipment.

BRITISH STANDARD

BSI EN365

BS EN
365:2004
Incorporating
corrigendum no. 1



DRAFT INTERNATIONAL STANDARD ISO/DIS 22846-2

ISO/TC 94/SC 1

Secretariat: SA

Voting begins on:
2010 11 11

Voting terminates on:
2011 01 11

INTERNATIONAL ORGANIZATION FOR STANDARDIZATION - INTERNATIONAL ORGANISATION DE NORMALISATION - ORGANISATION INTERNATIONALE DE NORMALISATION

ISO 22846-2

Notes: _____

3. Purchasing and certification of equipment:

- 'Purchasers should ensure that components in any system are compatible and that the safe function of any one component does not interfere with the safe function of another.' (ISO 22846-2 Sec. 7.1)
- When purchasing work at height equipment from a reputable dealer (such as Gravity Gear) an itemized invoice must be provided. On the invoice, details such as the date of purchase, items purchased and the manufacturer's article code numbers will be recorded.

NB: When equipment gets designed it needs to be done in a way for it to be practical and logical to use. Once the equipment has been designed it will be manufactured according to a specific standard and the grade of materials used is strong enough to withstand the necessary forces that it was designed for. It will then undergo internal testing to identify if there are any potential faults with the equipment, once all issues have been addressed it will be re-manufactured and sent for testing at a certified test centre. The test centre will test the equipment according to the specific EN standards it needs to comply with before it will be certified. When the certification of the equipment is done, the testing centre will issue a certificate of conformity to the manufacturer to prove that the equipment complies, then the manufacturer will be able to distribute/sell the equipment with a user and inspection manual.

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GEAR**



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- All equipment bought for and used in industrial work at height should comply with ISO 22846-2 Sec. 7 and EN 365.
- All work-at-height equipment must have a certificate stating:
 - a) to which standard the equipment was manufactured;
 - b) the equipment's safe working load or failure load; or
 - c) other certification that proves the equipment's reliability.

NB: The equipment controller has to ensure that all equipment purchased is loaded on a database to keep a record of the different types of equipment as well as the quantities of each item piece of equipment.

ITEM TYPE	ID.	SERIAL NO.	D.O.M:	P/M/R	INSPECTION DATE	INSPECTED BY	Date of purchase
FA HARNESS	GFHA29	141M001292121		P	19-Dec-22	M.Cilliers	2022.10.02
FA HARNESS	GFHA30	114M001292121		P	19-Dec-22	M.Cilliers	2022.10.02
FA HARNESS	DHA06	028M001292121		P	19-Dec-22	M.Cilliers	2022.09.15

FA HARNESS
FA HELMET
FIRST AID KIT
MAILLON
DELTA MAILLON
OMNI MAILLON
OVAL MAILLON
PROXI

- All work-at-height equipment should be supplied with an instruction manual specifying all the limitations and instructions for use, as well as the inspection criteria which must be strictly adhered to. A copy of the user/inspection manual should be given to the technician to study and understand as per international code of practice (ICOP 2.7.1.7.)

Notes: _____

4. Identification of equipment:

It is crucial for anyone working with work-at-height equipment, to be able to identify the equipment and to know what the equipment consists of, example: hardware, software or both.

		
Software	Both	Hardware

5. Marking and traceability of equipment:

'Equipment should be traceable to the relevant test certificates or certificates of conformity, and matched to records of use in order to facilitate its proper care.'(ISO 22846-2 Sec. 7.5 and EN 365 Sec. 4.8)

Equipment can be marked with different colour coding and with an individual ID number relevant to the company name to ensure easy traceability. for example, GTSA001 - (Company 'Gravity Training' - Shock Absorber - 001)

Software:

'All load-bearing access equipment should be marked with a unique identifier, ideally by the manufacturer, to allow traceability from manufacture to final disposal.'

'Items that have not been provided with a unique identifier by the manufacturer should be indelibly marked in a manner that does not affect their integrity. Particular care should be taken to ensure that webbing and rope products are not marked with damaging chemicals or potentially harmful adhesive products.'

- Mark equipment on tags.
- Ensure that markings do not cover critical inspection areas.
- Always adhere to manufacturer recommendations.
- Software can also be marked with a QR code labeling system, however, this system can be expensive.
- ***Do not mark software equipment on load-bearing components like the webbing or stitching.***



Hardware:

'Metal items should not be marked by stamping, unless by agreement with the manufacturer.'

- May be engraved, however, the engraving may not be deeper than 1 mm.
- May be marked with ink or paint.
- Ensure that markings do not cover critical inspection areas.
- Always adhere to manufacturer recommendations.
- Hardware can also be marked with a Blue dot technology that can make booking equipment in and out easier..
- ***Do not mark hardware on load-bearing/vital components like rivets, gates, springs etc.***



Ropes:

- May be marked similarly to how software is marked.
- Ensure both ends of ropes are marked.
- Only mark ends of ropes with labels.
- Ensure that no marking obscures any areas that may need to be inspected in future. if this is unavoidable, ensure marking methods are easily removable and replaceable.
- Ropes can be marked with a chip system , whereby a chip is inserted with a needle. The chip will contain all the details of the rope.
- Should a rope be damaged on the ends or in the middle, the whole rope does not necessarily have to be discarded. one can simply cut the damaged section out and split the rope in two. If this is done, remember to update your system and give the second rope its own identification number and label.



No marking at all

6. Records of equipment usage:

'Advice shall be given that a record is kept for each component, subsystem and system'. The record should contain headings for and spaces to allow entry of the following details, as laid out by the EN 365 Sec. 4.6 and ISO.

As per page five, the example of the inventory list is vital for record keeping. However, keeping records of equipment issued to a user (PPE) and equipment that is occasionally booked out (rigging, rescue and rope access equipment) also needs a form of traceability.

'Records should be kept to track the use of individual pieces of equipment.'(ISO 22846-2 Sec. 7.6)

These records should include at least the following:

- Unique identification marks.
- Date of entry into service.
- Storage location.
- Locations and periods of use.
- Abnormal uses or abnormal conditions in which equipment has been used.
- Any exposure to chemicals, abrasion or heavy grit.
- Any falls arrested or large loads imposed.
- Details of repairs or modifications.
- Any workmate retrieval carried out (rescue performed).
- The date, type and results of inspections for:
NB: Pre-use inspections.
Detailed inspections.
Interim inspections.

Typical user log, for team equipment

USER LOG — TEAM EQUIPMENT (SETS)

TEAM NUMBER _____

INVENTORY:

Item	No. Off	ID No.
Karabiners		
Mollies		
Descenders		
Backup devices		
Chest ascenders		
Sit harnesses		
Chest harnesses		

Item	No. Off	ID No.
Pulleys		
Slings		
Wire straps		
Seals		
Ascenders		
Rope protectors		

LOG:

Date taken	Destination	Taken by	Signature	Date returned	Signature	Comments

7. Retention period:

(ISO 22846-2 sec 7.7.1.3)

'Records shall be stored in a safe place for a minimum period of five years after the equipment has been withdrawn from service'.

8. Types of equipment inspections:

- **Pre-use check/inspections (ISO 22846-2 Sec. 7.7.2.1)**

An inspection that is both visual and tactile and that needs to be done before every use. Should any item show defects it must be withdrawn from service. Good practice requires this inspection to be documented by the user.

- **Detailed inspections (ISO 22846-2 Sec. 7.7.2.2)**

An inspection that has to be done by a competent person at intervals not exceeding six months, The equipment should be thoroughly inspected and a record of this inspection must be kept.

The competent person in CEIM should be able to understand the correct methods of operation and limitations of any equipment they may use.

- **Interim inspections (ISO 22846-2 Sec. 7.7.2.3)**

Is a further inspection that should be carried out where events liable to jeopardize safety (e.g. abnormal use, harsh conditions and heavy loading) have occurred. These are inspections over and above the normal pre-use check/inspection. These inspections should also be carried out by a competent person at intervals determined by the risk assessment, and should be recorded.

B. Inspection of equipment

1. Practical reasons for equipment inspections:

- Any damage that would compromise the normal working action/operation of the working-at-height equipment must be identified before use. There is a wide range of possible causes of degradation of materials in fall protection equipment, including but not limited to abuse, general wear and tear, edge/surface damage, ultra violet light, dirt, grit, chemicals, subjection to excessive loading and or falls. Textiles also deteriorate slowly with age regardless of the use.
- Equipment that has suffered a high shock load (impacted force), or that has been dropped from significant heights must be rejected and destroyed.
- The safest course of action is to reject any components about which there is any doubt.
- Load testing should not be carried out on any work-at-height equipment components. The requirements of the Driven Machinery Regulation does not apply to work-at-height equipment. Manufacturers of the equipment should supply you with a conformity certificate as proof that the equipment was tested by a test centre.



2. General rules of equipment inspections:

- Software should not be used beyond the lifespan given by the manufacturer.
- All equipment must be inspected at least every six months, and the details of the inspection must be recorded on a six-monthly full inspection form. (Please note that no white-out/tippex is allowed in the completion of an equipment inspection form as it compromises the reliability of the inspection form).
- Equipment should be inspected daily on site by the work team before access work commences. On long duration jobs, equipment should be subjected to a further weekly inspection and a weekly equipment inspection form should be completed.
- Items of equipment which are found to be sub-standard during these inspections should be marked as 'sub-standard' or 'rejected' and destroyed at the earliest opportunity. Details of why the equipment was rejected must be documented in the main inventory file.
- Items of equipment must be inspected as a unit as specified/indicated on EN standard item unless the manufacturer indicates in the user/inspection manual that the items need to be inspected individually.

Notes: _____

3. Examples of damaged equipment:





4. Storage of equipment:

'Information on storage of equipment should be provided by the manufacturer and this should be strictly followed, in addition to the information contained in this international standard.'(ISO 22846-2 Sec. 7.10)

- All equipment must be stored in a designated storeroom. The storeroom should, preferably, be a cool, dry and well ventilated place, where equipment is not exposed to sunlight, excessive heat, high humidity or corrosive substances.
- Access equipment storerooms should only be used for storing access equipment.
- Controlled access to the storeroom is very important. The storeroom, its maintenance and record keeping are the responsibility of the equipment controller.
- No equipment should rest on the floor, but equipment must be suspended from the walls, packed on suitable shelving or placed in suitable containers.

5. Care and cleaning of equipment:

'Information on cleaning and care of equipment should be provided by the manufacturer and this should be strictly followed, in addition to the information contained in this international standard.'(ISO 22846-2 Sec. 7.8)

- Equipment must be cleaned, washed, dried and prepared for storage.
- Equipment should only be washed with soap approved by the manufacturer. The soap should not contain any strong chemicals – soap used for equipment is similar to the soap used to clean ropes. Equipment made entirely from metal can be cleaned by submerging in clean, hot water containing detergent or soap.
- After washing the equipment, it should be rinsed in clean, cold water and dried naturally away from direct sunlight.
- No high-pressure steam cleaners should be used for cleaning of any equipment.



- No seawater should be used to wash equipment.
- If the equipment is contaminated by chemicals, it should be withdrawn from service immediately and the relevant original equipment manufacturer (OEM) should be contacted for advice.
- In the event where disinfectants are needed, advice should be sought from the manufacturer or supplier of the equipment about suitable disinfection agents.
- One has to consider two things when choosing a disinfectant: it should be effective in the combatting of diseases and the disinfectant should not create any adverse effects on the equipment. After disinfection, the equipment must be rinsed thoroughly in clean, cold water and dried naturally away from direct sunlight.

6. Maintenance:

'Information on maintenance of equipment should be provided by the manufacturer and this should be strictly followed, in addition to the information contained in this international standard.' (ISO 22846-2 Sec. 7.9)

Before one can carry out the manufacturer's instructions on equipment maintenance, one needs to know which requirements the manufacturer adhered to when producing the equipment. This is necessary to ensure that the manufacturer's instructions are in line with the relevant EN 365 standard (Sec. 4.3 and 4.5).

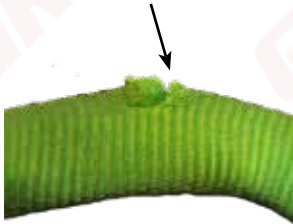
- Any maintenance of equipment should be carried out by using the manufacturer's instructions where available. There should be no deviation from these instructions without the prior approval of the manufacturer.
- Maintenance must be done by a competent person.
- Where necessary, the maintenance should be done by a representative from the manufacturer or by a third party specialist who has been approved by the manufacturer.
- Any item that cannot be repaired should be marked as unfit for service and be quarantined or discarded.
- Once equipment has been repaired, it should be inspected by a competent person before re-entry into service.

7. Manufacturer's guide about wear on hardware and software:

Most manufacturers have a general rule about the amount of wear that is acceptable on equipment before it is necessary to reject the equipment. How much wear equipment can withstand is different for hardware and software equipment.

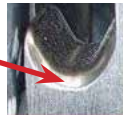
Software:

- Unfortunately, software does not have a wear indicator like hardware does. However, the rule for software is that no item should be used if it has wear or damage (e.g. regular wear, cuts, burns, exposure to chemicals) of more than 5%.



Hardware:

- Most hardware devices have a wear indicator that gives a guideline on when a device is no longer fit for use.
- Generally, however, the rule is that no wear on hardware should exceed a depth of 1 mm, regardless of the type of wear (e.g. regular wear, a cut or even engraving).
- Note that if there is any wear on bolts or rivets, the equipment should be rejected regardless of the depth of the wear.



8. Component inspection:

Every piece of work-at-height equipment consists of different components, and every component has inspection criteria which indicates how the component must be inspected..

Springs:

- Must be able to move freely in both directions
- No rust
- No bends
- Must be intact

Cams:

- Must be within the wear and tear limitations
- No rust
- No sharp edges
- Variable friction/pivot cams must be able to move freely
- Fixed cams must be secure

Bolts and rivets:

- No cracks, bends or deformities
- All rivets must be present
- No rust
- Must be secure

Webbing:

- No cuts or excessive abrasions
- No discolouration (e.g. from UV damage or chemical contaminants)
- Must be within the specified lifespan

Buckles:

- No cracks
- No rust
- Must be able to move freely
- No sharp edges

Length adjuster:

- No cracks
- No rust
- Must be able to move freely
- No sharp edges

Metal frames:

- No cracks or deformities
- No rust
- Must be able to move freely

Moving parts:

- No cracks
- No rust
- Must be able to move freely
- No sharp edges
- No strenuous sounds

Shock absorbers:

- No deployment
- Covering must be intact
- Connectors must be intact and in good working condition
- No cracks
- No rust
- Must be able to move freely
- No sharp edges
- No deformation

Plastic:

- No cracks
- No heat damage
- No UV damage
- No chemical damage
- No deformities

Ropes:

- No core breaks
- Internal core must not be visible
- No fraying
- No heat damage
- No chemical damage
- No excessive wear and tear

D-ring:

- No cracks or deformities
- No rust
- No excessive wear and tear

Notes: _____

9. Rejection criteria:

Herewith is a list of rejection criteria which states the various reasons why equipment could be rejected. If any of the below criteria matches the damage to the equipment, that is reason enough to reject the equipment.

Criteria number	Rejection criteria	Description
1.	Compromised stitching	Loose and damaged stitching compromises the strength of the item and could result in the item breaking, which could lead to serious injury or death.
2.	UV damage (Discolouration)	Equipment exposed to excessive sunlight can show signs of discolouration and stiffness. Once these signs are evident on the item, it should be discarded and replaced.
3.	Burns	Burns can occur due to cigarettes, open flames or even due to pieces of software that has been chafing against one another. Ensure that no person smokes near any rope access or fall arrest equipment. Once any signs of burns are evident, discard the item and replace it.
4.	Cuts	Cuts are a clear sign of an item weakening and should be treated as one of the most severe forms of damage. Should any cut or crack be seen on your equipment, it should be discarded and replaced immediately.

5.	Heavy soiling	Avoid standing on equipment or storing equipment on the ground. Soil creates micro chafing inside the equipment and can, therefore weaken the equipment. Custom rope soap can be used on your rope access or fall arrest equipment. Should heavy soiling not be removable, the item must be discarded and replaced.
6.	Chemical damage	Equipment exposed to any sort of chemical damage must be disposed of and replaced immediately. Types of chemical damage includes, but is not limited to, damage from pen markings, acids, paints, battery acid, glue, epoxy, blood and urine, etc. Signs of chemical damage is not always visible but could include discolouration and stiffness. When load-bearing items are marked, ensure that a rope marker is used and not a permanent marker.
7.	Outdated equipment (Date of manufacture)	According to industry best practice, software items can be used for up to ten years after the manufacture date. However, it is recommended to always follow the manufacturer's guidelines in this regard.
8.	Wear and tear	Wear and tear of equipment can usually be seen through signs of cracks, burns, cuts, discolouration, loose threads or bends or the manufacturer limitation of the equipment being exceeded. A load or wear indicator can usually be found on your equipment, which marks what the limit of use for the equipment is. Once any signs of wear and tear are evident, the item must be discarded and replaced.

9.	Hairline and deep cracks	Cracks are a clear sign of an item weakening and should be treated as one of the most severe forms of damage. Should any crack be seen on an equipment item, the item should be discarded and replaced with new equipment.
10.	Impaired functionality	A functionality test must always be done on equipment to ensure that the equipment is functioning as smoothly as it did when it was first bought. Equipment that is no longer functioning as intended should be rejected. Functionality of equipment must always be done on your equipment. The locking system of an item must always be fully functional and should not creep under load.
11.	Rust and corrosion	Special care must be taken when inspecting rust and corrosion on items. Surface rust can still be accepted but deep corrosion will weaken the item. Also note that some hardware items can be used in conjunction with software and can therefore be damaged by rust, e.g ropes and descenders.
12.	Compromised integrity of bolts, rivets and spring loading	If bolts, rivets or spring loading items are no longer intact, the quality of the equipment item has been compromised, and the equipment will no longer function properly. Should any part of the equipment be out of place, the equipment should be discarded and replaced with new equipment.

13.	Compromised buckles working action	Buckles can often be loaded incorrectly resulting in incorrect use of equipment. This can usually be fixed but it is recommended that the user of the equipment (e.g, harness, work belt, shock absorber) be sent for retraining.
14.	Internal damage and deployment	If any hard or soft spots are observed on a rope it means that the rope has been damaged internally, and it should no longer be used. Even if no external damage is visible in this case, the internal damage renders the rope unusable. Harnesses have fall indicators that deploy during a high shock load. Once these fall indicators have deployed, the item should no longer be used. Shock absorbers that have been deployed have experienced serious shock load and are, therefore, unable to absorb any further shock. Such items are no longer safe for use and should be discarded immediately.
15.	Bending	If equipment is used incorrectly or placed under tremendous stress, the hardware can bend and even break. Bending of hardware weakens the item and compromises its integrity. A common example of this is when pylon hooks (scaffold hooks) are cross-loaded and force is applied. In such cases, the item will bend and oftentimes break.

Notes: _____

10. List of work-at-height PPE

Helmet (EN12492/397)

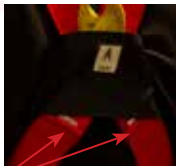
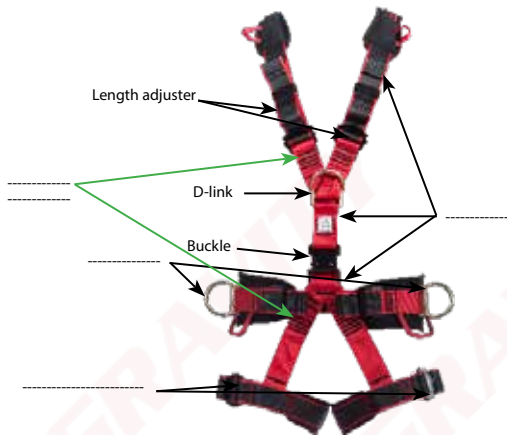
Description and uses:

- A climbing helmet serves to guard a worker's head from possible side impact forces while working.
- Climbing helmets are fitted with a chin strap to ensure the helmet stays on the worker's head at all times.



Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.		
1. Compromised stitching	6. Chemical damage	11. Rust and corrosion
2. UV damage (discolouration)	7. Outdated equipment (Date of manufacture)	12. Compromised integrity of bolts, rivets and spring loading
3. Burns	8. Wear and tear	13. Compromised buckles working action
4. Cuts	9. Hairline and deep cracks	15. Bending
5. Heavy soiling	10. Impaired functionality	

Full body harness (EN 361, 358, 813)



Fall indicator

Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.

1. Compromised stitching	6. Chemical damage	11. Rust and corrosion
2. UV damage (discolouration)	7. Outdated equipment (Date of manufacture)	12. Compromised integrity of bolts, rivets and spring loading
3. Burns	8. Wear and tear	13. Compromised buckles working action
4. Cuts	9. Hairline and deep cracks	14. Internal damage and deployment
5. Heavy soiling	10. Impaired functionality	15. Bending

Description and uses:

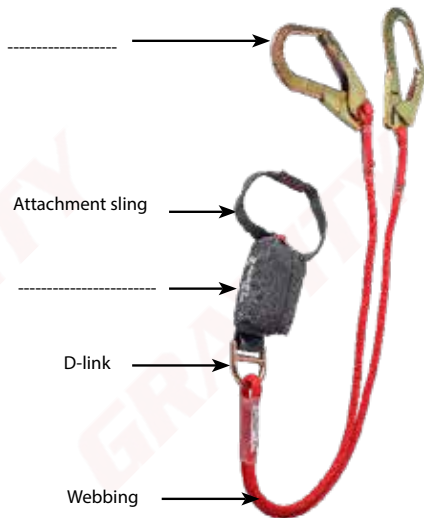
- A harness is a worker's safe connection point into a fall arrest or work positioning system.
- A harness has dedicated work positioning and fall arrest attachment points: the top D-link (fall arrest) and the side D-links (work positioning).

Shock absorbing lanyard (EN 355)

Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.		
1.	6. Chemical damage	11.
2. UV damage (discolouration)	7.	12. Compromised integrity of bolts, rivets and spring loading
3.	8. Wear and tear	13.
4. Cuts	9.	14. Internal damage and deployment
5.	10. Impaired functionality	15.

Description and uses:

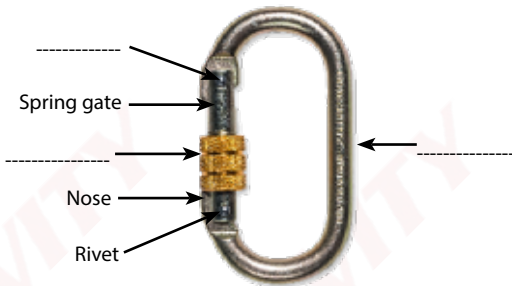
- A shock absorbing lanyard is a fall arrest system that is used to safely stay connected to the structure at all times, while working or climbing.



Connectors (EN 362)

Description and uses:

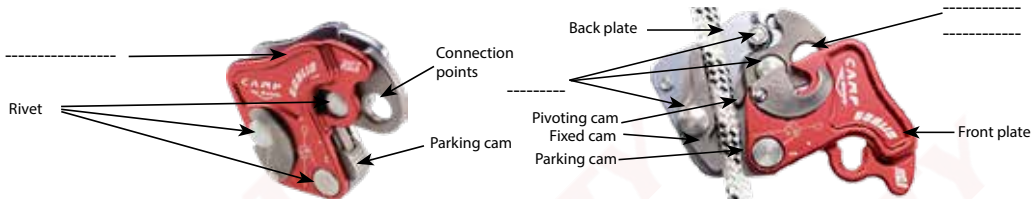
- Used to connect different types of equipment.
- An example of a connector is a karabiner. There are two types of karabiners: a twist lock karabiner and a screw gate karabiner.
- It is important to always load a karabiner in its length. DO NOT CROSS-LOAD a karabiner.



Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.

4. Cuts	11. Rust and corrosion
6. Chemical damage	12. Compromised integrity of bolts, rivets and spring loading
8. Wear and tear	15. Bending
9. Hairline and deep cracks	
10. Impaired functionality	

Rope grab (EN 353-2)



Description and uses:

- This device clamps onto the rope and is used as a fall arrest device on a pre-rigged vertical lifeline.

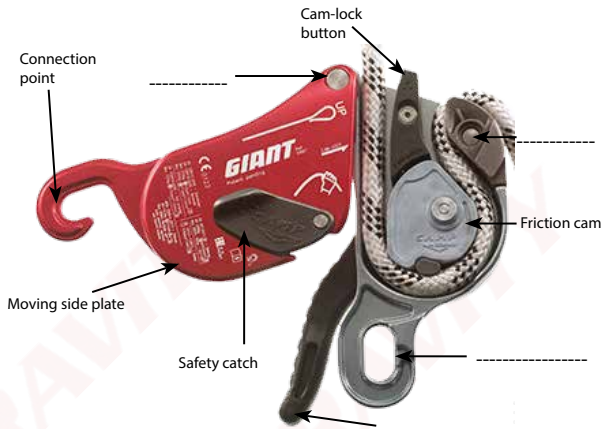
Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.	
4. Cuts	
8. Wear and tear	15. Bending
9. Hairline and deep cracks	

Descenders

Camp Giant (EN 341, 12841-C)

Description and uses:

- This device has variable friction that allows descending on a rope at variable speeds.
- Used as a lowering component in a rescue system.
- Used to tension a horizontal lifeline.



Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.

	11. Rust and corrosion
6. Chemical damage	12. Compromised integrity of bolts, rivets and spring loading
	15. Bending

Pulleys (EN 12278)

Description and uses:

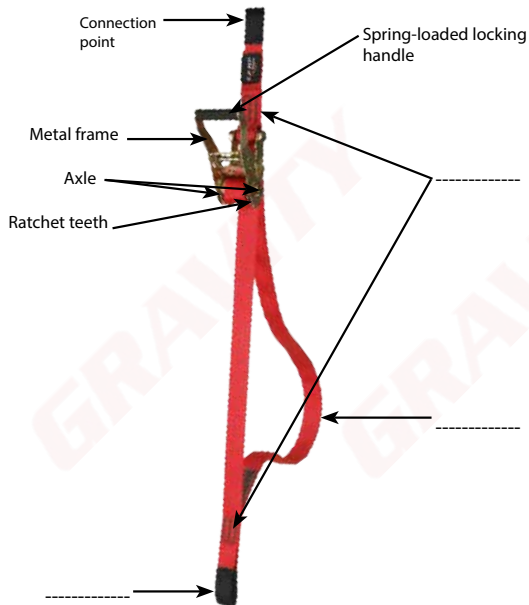
- Pulleys are wheels that carries a flexible rope, cord, cable, chain, or belt on its rim. Pulleys are used singly or in combination to transmit energy and motion. A load is attached to the other end of the rope or to the pulley itself. The operator pulls up or down on the other end of the rope. Pulleys with grooved rims are called sheaves. The pulley reduces the friction on the rope (bigger pulley = less friction; smaller pulley = more friction).
- It's easier to pull something down than to lift it straight up; the pulley thus changes the direction of pull to enable you to lift your load with an easier direction of pull.



Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.

4. Cuts	
	12. Compromised integrity of bolts, rivets and spring loading
8. Wear and tear	
10. Impaired functionality	

Rescue ratchet sling (EN 354)



Description and uses:

- Used to transfer weight of casualty off fall arrest equipment.

Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.

1. Compromised stitching	
	9. Hairline and deep cracks
3. Burns	
	11. Rust and corrosion
5. Heavy soiling	
	15. Bending
7. Outdated equipment (Date of manufacture)	

Rope (EN 1891 or EN 892)

Description and uses:

- There are two types of ropes: semi-static ropes (most often used in work-at-height; has a stretch ability of 5–10%) and dynamic ropes (has a stretch ability of 10–25%, and can thus withstand higher shock loads).
- Work rope is made of an outer sheath and an inner core. The outer sheath protects the inner core and equals only 20% of the strength of the rope. The inner core is where 80% of the strength of the rope lies.
- **NB:** Any edge with a radius of less than 5 mm is considered a sharp edge. Always ensure that, when working with ropes, any sharp edges are covered with edge protection. This ensures personal safety and prolongs the lifetime of the rope.



Identify the rejection criteria/damage that can cause this item to be rejected, as per Section 8.

2. UV damage (discolouration)

3. Burns

4. Cuts

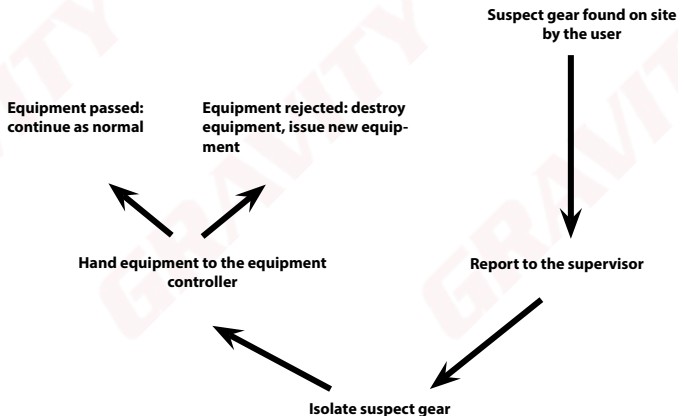
8. Wear and tear

14. Internal damage and deployment

11. Reporting suspect equipment.

Equipment that is found to be damaged or too old for use must be reported to the person responsible for the site for example, the site supervisor or safety officer.

Damaged or old gear must be properly marked and removed from service immediately, to be destroyed at the earliest opportunity to prevent any possible incidents.



NB: Equipment may only be repaired by either the manufacturer or a company certified to do so by the manufacturer. **NO CONTRACTOR MAY REPAIR PPE.**

C. Examples of inspection registers and class exercises.

Example of a pre-use inspection register

PPE INSPECTION SHEET					
No:	Kit Number	Safe to use? Y/N	User (Responsible Person)	Signature	Date
1					
2					

Examples of a detailed inspection register.

Refer to rejection criteria (pg. 20) to indicate why an item could be monitored or rejected and complete as a class exercise.

Item Type.	EN standard	ID No or Serial No	D.O.M	Results. (P/M/R)	Comments:
Connector					
Harness					
Descender					
Ropes					
Slings					
Ascender					
Pulley					
Shock-absorber					
Work positioning lanyard					
Helmet					

Inspected by: _____

Signature: _____

Inspection date: _____

Use Section 8, page 21–24, to complete the below tables regarding rejection criteria for hardware and software equipment.

Complete as a class exercise

Identify nine possible rejection criteria that could be found on software equipment.		

Identify ten possible rejection criteria that could be found on hardware equipment	

Annexure 1

Rejection criteria

Herewith is a list of rejection criteria which states the various reasons why equipment could be rejected. If any of the below criteria matches the damage to the equipment, that is reason enough to reject the equipment.

Criteria number	Rejection criteria	Description
1.	Compromised stitching	Loose and damaged stitching compromises the strength of the item and could result in the item breaking, which could lead to serious injury or death.
2.	UV damage (Discolouration)	Equipment exposed to excessive sunlight can show signs of discolouration and stiffness. Once these signs are evident on the item, it should be discarded and replaced.
3.	Burns	Burns can occur due to cigarettes, open flames or even due to pieces of software that has been chafing against one another. Ensure that no person smokes near any rope access or fall arrest equipment. Once any signs of burns are evident, discard the item and replace it.
4.	Cuts	Cuts are a clear sign of an item weakening and should be treated as one of the most severe forms of damage. Should any cut or crack be seen on your equipment, it should be discarded and replaced immediately.

5.	Heavy soiling	Avoid standing on equipment or storing equipment on the ground. Soil creates micro chafing inside the equipment and can, therefore weaken the equipment. Custom rope soap can be used on your rope access or fall arrest equipment. Should heavy soiling not be removable, the item must be discarded and replaced.
6.	Chemical damage	Equipment exposed to any sort of chemical damage must be disposed of and replaced immediately. Types of chemical damage includes, but is not limited to, damage from pen markings, acids, paints, battery acid, glue, epoxy, blood and urine, etc. Signs of chemical damage is not always visible but could include discolouration and stiffness. When load-bearing items are marked, ensure that a rope marker is used and not a permanent marker.
7.	Outdated equipment (Date of manufacture)	According to industry best practice, software items can be used for up to ten years after the manufacture date. However, it is recommended to always follow the manufacturer's guidelines in this regard.
8.	Wear and tear	Wear and tear of equipment can usually be seen through signs of cracks, burns, cuts, discolouration, loose threads or bends or the manufacturer limitation of the equipment being exceeded. A load or wear indicator can usually be found on your equipment, which marks what the limit of use for the equipment is. Once any signs of wear and tear are evident, the item must be discarded and replaced.

9.	Hairline and deep cracks	Cracks are a clear sign of an item weakening and should be treated as one of the most severe forms of damage. Should any crack be seen on an equipment item, the item should be discarded and replaced with new equipment.
10.	Impaired functionality	A functionality test must always be done on equipment to ensure that the equipment is functioning as smoothly as it did when it was first bought. Equipment that is no longer functioning as intended should be rejected. Functionality of equipment must always be done on your equipment. The locking system of an item must always be fully functional and should not creep under load.
11.	Rust and corrosion	Special care must be taken when inspecting rust and corrosion on items. Surface rust can still be accepted but deep corrosion will weaken the item. Also note that some hardware items can be used in conjunction with software and can therefore be damaged by rust, e.g ropes and descenders.
12.	Compromised integrity of bolts, rivets and spring loading	If bolts, rivets or spring loading items are no longer intact, the quality of the equipment item has been compromised, and the equipment will no longer function properly. Should any part of the equipment be out of place, the equipment should be discarded and replaced with new equipment.

13.	Compromised buckles working action	Buckles can often be loaded incorrectly resulting in incorrect use of equipment. This can usually be fixed but it is recommended that the user of the equipment (e.g, harness, work belt, shock absorber) be sent for retraining.
14.	Internal damage and deployment	If any hard or soft spots are observed on a rope it means that the rope has been damaged internally, and it should no longer be used. Even if no external damage is visible in this case, the internal damage renders the rope unusable. Harnesses have fall indicators that deploy during a high shock load. Once these fall indicators have deployed, the item should no longer be used. Shock absorbers that have been deployed have experienced serious shock load and are, therefore, unable to absorb any further shock. Such items are no longer safe for use and should be discarded immediately.
15.	Bending	If equipment is used incorrectly or placed under tremendous stress, the hardware can bend and even break. Bending of hardware weakens the item and compromises its integrity. A common example of this is when pylon hooks (scaffold hooks) are cross-loaded and force is applied. In such cases, the item will bend and oftentimes break.

01. *THE WAY FORWARD*



Congratulations on completing Gravity Training's Climbing Equipment Inspection & Management course.

If found not yet competent (NYC), you will need more training before you can be reassessed. Please do not be discouraged. Not everyone passes on their first attempt. Some learners need a bit more experience to obtain the required knowledge and skills. You will receive comprehensive feedback on your performance and be informed on how to proceed.

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